

Growing the 3D coverage — methods that compound over time

Every method is grounded: dimensions come from FLIS or the manual's own drawings, never invented. Coverage climbs as OCR, FLIS refreshes, and curation accrue.



THE GROWTH LADDER (grounded → progressively richer)

1 · Expand the FLIS dimension parser

Use the fields already in your data: 'overall length', 'body diameter', 'nominal thread size', radius, depth. Prefer OVERALL dims for accuracy; gate the button to renderable parts.

Grounded: data already indexed · pushes ~20.9k → ~30k+ · low effort

FREE · NOW

2 · Parametric family templates

For standard families (bolt, nut, washer, flange, connector, bearing) render a real shape from the characteristics + thread — not just a box. Recognizable, still grounded.

Grounded: standards + FLIS dims · med effort

RICHER SHAPES

3 · Extract dimensions from the TM drawings

Read dimension callouts off the figure/exploded-view pages (OCR boxes + dimension-line detection), tie them to the figure's part via the parts index, attach as cited dims. Literally 2D sketch → 3D dims.

Grounded: the manual's own drawing, cited · needs OCR done · med-high effort

YOUR ASK

4 · Multi-view orthographic reconstruction

Where a TM gives front/side/section dimensioned views, reconstruct a more accurate solid (real engineering reconstruction).

Grounded: the drawing's own views · high effort

HIGHER FIDELITY

5 · Real CAD / photogrammetry (look-alikes)

For the highest-value look-alike parts, import real CAD or photograph the part — exact surfaces where it matters. A library that grows.

Grounded: real geometry · per-part labor / data-rights

EXACT, SELECTIVE

6 · SME confirm / curate loop

Let an SME confirm/correct a part's dims or attach a model; stored & reused (like the parts learning). Coverage grows with use.

Grounded: human-verified · low effort, ongoing

LEARNS

WHY IT COMPOUNDS

Each OCR pass makes more drawings readable (feeds #3/#4); each monthly FLIS refresh adds characteristics; each parser/template/SME addition lifts the renderable count — all additive (R1) and append-only (R6). A per-vehicle '3D coverage %' meter makes the climb visible. The representative model is the floor; CAD/photogrammetry is the ceiling for the parts that earn it.